

SAGE: SRL Conference

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Abstract—SRL ABS

Index Terms—

I. INTRODUCTION

The rapid adoption of Large Language Models (LLMs) has made scalable, conversational tutoring practical, yet their default behavior, i.e., producing complete and fluent answers to user prompts, sits uneasily with how learners actually acquire competence. The productive struggle of recalling, attempting, and self-correcting is precisely what an eager answer-giver removes [?], [?]. Effective instruction is therefore *regulated*. It begins as Instructor-Regulated Learning (IRL), in which instructors plan, sequence, and monitor what a learner engages with, and aims ultimately at Self-Regulated Learning (SRL), in which learners set their own goals, monitor their own comprehension, and evaluate their own progress [?], [1].

Our prior work addressed the instructor’s side of this arc. SAGE [?] enforces instructional policy at the retrieval boundary through a three-layer Safety Gatekeeper, and its *cognitive* learner model [?] earns rather than assumes the set of mastered topics on which that gate acts, tracing several kinds of evidence and certifying a concept only when all of them are demonstrated. Herein lies a limitation, however. That system regulates capably on the instructor’s behalf, i.e., it governs what a learner may access and certifies what they have mastered, but it treats the learner as the *object* of regulation rather than a participant in it, since the learner’s own judgment of what they know never enters the model.

At the heart of self-regulated learning is exactly that judgment, viz. the learner’s monitoring and self-evaluation of their own understanding [1]. A tutor that ignores it forfeits the learner’s most direct signal of where they are struggling. Yet a tutor that simply trusts it invites a well-documented failure mode, as learners are systematically overconfident and tend to overestimate what they have mastered [?]. Open learner models, which expose the system’s estimate and allow the learner to contest it [?], face precisely this tension: permitting a learner to raise their own mastery estimate reintroduces the over-crediting that a competency gate exists to prevent. Crucially, this is the question we address, i.e., how a learner’s self-assessment can inform the knowledge model without letting overconfidence inflate it.

Our answer rests on an asymmetry. A learner who reports that they do not really understand a concept, i.e., that a correct answer was a lucky guess, supplies genuine evidence

of non-mastery; a learner who asserts mastery supplies none, since demonstrating mastery is what assessment is for. Towards this, we incorporate self-assessment in a *downward-only* manner: it may lower the model’s confidence and demand further evidence, but it may never raise that confidence or shortcut certification. We build this asymmetry into a multidimensional learner model that extends the cognitive model of [?] with the learner’s metacognitive, affective, and motivational state, inferred from ordinary interaction and fed back into the tutor’s behavior. At its center is a *calibration loop* in which sustained disagreement between a learner’s self-assessment and the model’s estimate adapts the model’s per-learner guess parameter, thereby reshaping how future evidence is weighed rather than merely the score displayed on screen, under a guarantee that self-report can delay but never accelerate certification. Concretely, this paper makes the following contributions:

- We extend the cognitive learner model of [?], which tracks knowledge and skill for competency certification, into a *multidimensional learner model* that additionally infers a learner’s metacognitive, affective, and motivational state from interaction telemetry and couples it to an LLM tutor’s responses. We integrate established constructs for these dimensions rather than claiming them individually as novel; the contribution is their coupling to a knowledge-tracing tutor in service of self-regulation.
- We introduce the *SRL-BKT calibration loop*, in which sustained self-assessment disagreement drives downward-only adaptation of the per-learner guess parameter, allowing a learner’s own metacognition to recalibrate the knowledge model itself rather than only a displayed score.
- We prove a *no-self-certification guarantee*, i.e., that learner self-report can delay but never accelerate certification, so that the evidence-diversity guarantee of the underlying model is preserved intact.
- We evaluate the system in a *semester-long classroom deployment* [PLACEHOLDER: ~ 3 months, $N \approx 50\text{--}70$ students, IRB #...], reporting whether the loop fires often enough to matter, whether it improves the model’s calibration against independent performance, and how it relates to self-regulation outcomes in authentic use.

The classroom deployment is central to the paper’s claim rather than an afterthought. A calibration mechanism can be sound in principle yet inert in practice, i.e., firing too

rarely to change anything, or improving only self-agreement rather than agreement with real competence. Towards this, evaluating over a full semester of authentic coursework allows us to test the mechanism where those risks actually surface.

The remainder of this paper is organized as follows. Section II reviews self-regulated learning support, open learner models, and metacognitive calibration. Section III recaps the underlying cognitive knowledge-tracing model and specifies the multidimensional learner model and the calibration loop, together with the no-self-certification guarantee. Section IV reports the classroom deployment, and Section V concludes.

II. RELATED WORK

A. Supporting Self-Regulated Learning

Self-regulated learning is not a fixed trait but a cyclical, learnable process, viz. the forethought, performance, and self-reflection phases through which a learner sets goals, monitors strategy use, and evaluates outcomes [?], [?]. Intelligent tutoring systems have long sought to support this process, i.e., through metacognitive prompts, help-seeking scaffolds, and reflective feedback that encourage learners to plan and monitor their own work [?], [1], and more recent work extends such scaffolding to LLM-based tutors by adapting the granularity of guidance to a learner’s inferred state [?]. Herein, however, lies a recurring pattern: these systems perform the monitoring and evaluation on the learner’s behalf, i.e., the system tracks what the learner appears to know and decides when to intervene, while the learner’s own assessment of their understanding is elicited, if at all, only to prompt reflection rather than to inform the model itself.

Beyond cognition, tutoring systems increasingly model the affective and motivational dimensions of learning, since a learner’s frustration, boredom, engagement, and goal orientation shape both their behavior and their outcomes [?], [2]. Towards a holistic learner representation, we likewise integrate metacognitive, affective, and motivational signals alongside the cognitive model. We emphasize, however, that we claim no novelty for these constructs individually; our contribution is their coupling to a knowledge-tracing tutor in the service of self-regulation, and in particular the mechanism, described next, by which a learner’s own self-assessment is permitted to act upon the knowledge model.

B. Open Learner Models and Learner Overconfidence

Exposing a learner model to the learner, rather than hiding it, is the province of open learner models (OLMs), which externalize the system’s estimate of what a learner knows so that the learner may inspect and, in negotiated variants, contest it [?]. The purpose is metacognitive, i.e., seeing one’s own estimated mastery prompts reflection and supports self-monitoring. Crucially, however, negotiated OLMs generally permit adjustment in either direction, and this is where a well-documented difficulty arises: learners are systematically overconfident, tending to overestimate what they have

mastered, and their judgments of their own learning are frequently poorly calibrated [?], [?]. Permitting a learner to raise their own mastery estimate therefore reintroduces precisely the over-crediting that a competency-gated tutor exists to prevent.

Our approach retains the OLM insight that a learner’s self-assessment is valuable, but departs from it in two respects. First, whereas an OLM adjusts the *displayed* estimate for the learner’s reflection, we feed the learner’s self-assessment into the *inference* itself, viz. into the parameter governing how much future evidence counts, so that the signal changes what the model will conclude rather than only what it shows. Second, to avoid the overconfidence trap, we admit self-assessment in a downward-only manner, i.e., a learner may lower the model’s confidence but never raise it, an asymmetry we justify and formalize in Section III.

C. Adapting Knowledge Tracing to the Individual

That a single set of knowledge-tracing parameters cannot fit every learner is well recognized. Individualized Bayesian Knowledge Tracing fits per-student parameters, e.g., a learner-specific prior or learning rate, so that the model reflects how quickly a particular student acquires a skill [?], [?], and neural knowledge tracers pursue a similar personalization implicitly by conditioning on a learner’s full response history. Herein, however, the adaptation is driven entirely by the learner’s *response data*, i.e., by further cognitive evidence of the same kind the model already consumes, and it may move an estimate in either direction. In contrast, the adaptation we introduce is driven by a *metacognitive* signal, viz. the learner’s own reported disagreement with the model; it is confined to the guess parameter; and it is constrained to be non-inflating by construction, such that it can slow certification but provably never hasten it.

D. Summary of the Gap

Table I situates these strands. Each supplies part of what a self-regulated, competency-gated tutor requires, but none supplies the whole. Open learner models let a learner contest an estimate, but adjust a display rather than the inference and place no principled limit on upward revision. Individualized knowledge tracing adapts inference to the individual, but from response data rather than the learner’s own metacognition, and without a guarantee against inflation. And the self-regulation scaffolds of intelligent tutors elicit a learner’s reflection without letting that reflection act upon the knowledge model at all. The calibration loop of this paper brings these together, i.e., it lets a learner’s sustained self-assessment adapt the inference itself, in a downward-only manner, under a guarantee that self-report can delay but never accelerate certification.

TABLE I: Where learner self-assessment can act, across representative approaches. ✓: supported; ×: not supported. Our method combines the learner-contestability of open learner models with the per-learner adaptation of individualized knowledge tracing, and adds a metacognitively driven, non-inflating adaptation with a certification-preserving guarantee.

Capability	OLM	Indiv. BKT	Ours
Estimate contestable by learner	✓	×	✓
Learner input adapts the inference	×	×	✓
Per-learner parameter adaptation	×	✓	✓
Non-inflating (down-only) by construction	×	×	✓
Certification-preserving guarantee	×	×	✓

III. METHODOLOGY

A central challenge in moving from instructor-regulated to self-regulated tutoring is that a learner’s own judgment of their understanding is at once the most direct signal of where they are struggling and the least trustworthy, i.e., learners are systematically overconfident and tend to overestimate what they have mastered. This section specifies a learner model that admits that judgment while structurally denying it the power to inflate certified mastery. We first recap the cognitive knowledge-tracing model on which we build (Section III-A), then extend it into a multidimensional model spanning the learner’s metacognitive, affective, and motivational state (Section III-B). At its center is a calibration loop through which a learner’s sustained self-assessment adapts the model’s inference itself (Section III-C), which we prove can delay but never accelerate certification (Section III-D). Finally, we describe how the inferred state conditions the tutor’s behavior (Section III-E).

A. Recap: The Cognitive Knowledge-Tracing Model

This paper builds directly on the cognitive learner model of [?], which we recap only to the extent the calibration loop requires. For each topic t , that model tracks three tier posteriors $\tilde{P}_t^{(k)} \in [0, 1]$ ($k \in \{1, 2, 3\}$ for declarative, procedural, and applied understanding), each updated online by Bayesian Knowledge Tracing as the learner works. On a correct response at tier k , the posterior is revised as

$$\tilde{P}_t^{(k)} \leftarrow \frac{\tilde{P}_t^{(k)}(1 - p_S^{(k)})}{\tilde{P}_t^{(k)}(1 - p_S^{(k)}) + (1 - \tilde{P}_t^{(k)})p_G^{(k)}}, \quad (1)$$

followed by a learning transition, where $p_S^{(k)}$ and $p_G^{(k)}$ are the slip and guess rates. A topic is *certified* ($t \in K$) only when every tier satisfies both a confidence and a minimum-evidence condition at once, i.e., $\forall k : \tilde{P}_t^{(k)} \geq \theta_{\text{cert}} \wedge N_t^{(k)} \geq N_{\text{min}}$. Two properties of Eq. (1) matter for what follows. First, the guess rate $p_G^{(k)}$ is a *rate* on evidence, i.e., a larger $p_G^{(k)}$ makes each correct response raise the posterior by less, since a higher assumed probability of guessing discounts the informativeness of a correct answer. Second, certification is evaluated on the raw posteriors $\tilde{P}_t^{(k)}$; any quantity presented to the learner is a separate, display-only transform. The calibration loop introduced below acts through exactly these two properties, viz. it adapts $p_G^{(k)}$ for an individual learner,

and it does so without ever writing to the raw posterior on which certification depends.

B. The Multidimensional Learner Model

The cognitive model above answers what a learner *knows*. To support self-regulation, we extend it into a multidimensional learner model that additionally represents how the learner monitors, feels about, and is motivated in their work, i.e., its metacognitive, affective, and motivational dimensions. The metacognitive dimension captures the learner’s own assessment and regulation of their understanding, viz. their reported confidence, the character of their help-seeking, and their adherence to a recommended path. The affective dimension tracks engagement-relevant emotion inferred from interaction, e.g., a frustration trajectory and an estimate of whether task difficulty matches skill. The motivational dimension reflects the learner’s goals and their alignment with the current activity. Each dimension is maintained as a read-only aggregation over the same interaction telemetry the cognitive model consumes, and together they drive a perception–inference–adaptation cycle in which the tutor’s behavior is conditioned on the inferred state (Section III-E).

We reiterate that the individual constructs comprising these dimensions are established, and we claim no novelty for them in isolation. The contribution of this paper is the coupling of one specific metacognitive signal, viz. the learner’s own disagreement with the model’s mastery estimate, back into the cognitive inference itself, to which we now turn.

C. The SRL–BKT Calibration Loop

Herein lies the central mechanism of this paper. The learner is able to express disagreement with the model’s estimate of their mastery through a downward-only control, i.e., they may signal that they know a concept less well than the model currently believes, but not more. Such an expression is treated as a *metacognitive observation* and processed in two layers.

Layer 1: effective mastery for adaptation. For display and for conditioning the tutor’s responses, but *not* for certification, the system forms an effective mastery that blends the model’s estimate with the learner’s self-assessment,

$$P_{\text{eff}} = \alpha P_{\text{BKT}} + (1 - \alpha) P_{\text{self}}, \quad \alpha = 0.6. \quad (2)$$

This lets a learner’s stated uncertainty immediately soften the tutor’s behavior, e.g., restoring scaffolding on a topic the model had considered strong, without touching the certified record.

Layer 2: parameter adaptation. A single self-report is weak evidence and may reflect a passing mood; a *sustained* pattern of it is not. We therefore track the direction of a learner’s adjustments with an exponential moving average,

$$D \leftarrow \lambda_{\text{cal}} d + (1 - \lambda_{\text{cal}}) D, \quad (3)$$

where d records the direction of each adjustment. When a learner has downgraded repeatedly and this average crosses

a threshold, i.e., at least $N \geq 3$ downgrades with $D < -0.3$, the system increases that learner's per-tier guess rate by a fixed step, subject to a hard ceiling,

$$p_G^{(k)} \leftarrow \min(p_G^{(k)} + \beta, p_{G,\max}^{(k)}), \quad \beta = 0.10, \quad (4)$$

with per-tier ceilings $p_{G,\max}^{(1,2,3)} = (0.40, 0.25, 0.15)$ for the declarative, procedural, and applied tiers; adaptation halts once the ceiling is reached. By Eq. (1), a larger $p_G^{(k)}$ makes each subsequent correct response raise the posterior by less. The persistent effect of sustained self-reported overconfidence, then, is that the model demands *more* correct evidence from that learner before it will certify, viz. the learner's own metacognition recalibrates how the model weighs their future performance, rather than editing a number on screen. The ceilings are chosen to remain strictly below the non-degeneracy boundary $p_S^{(k)} + p_G^{(k)} < 1$ at every tier (the tier sums are 0.50, 0.40, and 0.35), a property we rely on below.

D. The No-Self-Certification Guarantee

The calibration loop deliberately gives the learner influence over the inference. It is therefore essential to establish that this influence cannot be turned into a shortcut to certification. We show a two-sided safety property: self-report can only ever delay certification, never hasten it, and it can never render a learner's correct evidence worthless.

Proposition 1 (No self-certification). *Fix a learner and a topic t , and consider any sequence of subsequent responses. Relative to the same sequence processed without self-report, the learner's downward-only self-assessment (i) cannot cause topic t to become certified at an earlier response, and cannot cause a topic that would not otherwise be certified to become certified; and (ii) never drives any tier's update to a degenerate state, i.e., a correct response always raises the corresponding posterior. Consequently, the evidence-diversity guarantee of [?] holds unchanged under the calibration loop.*

Proof. Certification depends only on the raw posteriors $\tilde{P}_t^{(k)}$ and the evidence counts $N_t^{(k)}$ (Section III-A). Self-report acts on the system in exactly two ways, and neither writes to these quantities directly. Layer 1 modifies only P_{eff} (Eq. 2), which drives display and response adaptation but is excluded from certification by construction. Layer 2 modifies only the guess rate, and by Eq. (4) it can only *increase* $p_G^{(k)}$, up to the ceiling $p_{G,\max}^{(k)}$, never decrease it.

For (i), observe from Eq. (1) that the post-update posterior is strictly decreasing in $p_G^{(k)}$: for a fixed prior $\tilde{P}_t^{(k)}$ and a correct response, a larger $p_G^{(k)}$ yields a smaller updated posterior. Hence at every step the posterior under self-report is no larger than the posterior for the same response sequence without it, while the counts $N_t^{(k)}$ are identical in both cases. The certification condition $\tilde{P}_t^{(k)} \geq \theta_{\text{cert}}$ can therefore be reached no earlier under self-report, and never reached when it would not otherwise be; the counts

underlying the evidence-diversity guarantee are untouched, so that guarantee is preserved.

For (ii), the guess rate is bounded by $p_{G,\max}^{(k)}$, chosen so that $p_S^{(k)} + p_{G,\max}^{(k)} < 1$ at every tier. Under this condition the correct-response update in Eq. (1) is strictly increasing in $\tilde{P}_t^{(k)}$ and maps a correct observation to a strictly larger posterior, so no attainable value of $p_G^{(k)}$ renders correct evidence non-diagnostic. Self-report can thus slow a learner's certification but can never foreclose it. $\square$

The asymmetry this formalizes is epistemic, not merely a safety heuristic. A learner's admission that a correct answer was a guess is genuine evidence of non-mastery and is acted upon; a learner's assertion of mastery is not evidence, since demonstrating mastery under observation is precisely what the assessment provides. Clause (ii) further ensures the mechanism is corrective rather than punitive: a learner who under-reports persistently is asked for more evidence, not locked out, since their correct work continues to move the model. The calibration loop thus admits the informative half of a learner's self-assessment while structurally refusing the half that would invite over-crediting.

E. Metacognition- and Affect-Conditioned Adaptation

Beyond the calibration loop, the inferred state conditions how the tutor responds, closing the perception–adaptation cycle. When the affective dimension indicates sustained frustration, the tutor enters a reframing mode that suppresses raw code in favor of plain-language reassurance and analogy, i.e., it prioritizes re-engagement over content delivery. When interaction timing indicates rushing, the tutor shifts to a more concise mode that matches the learner's pace. And a learner-state safety filter conditions the permissiveness of the tutor on the learner's inferred state, e.g., withholding complex generated code when estimated cognitive load is high. These adaptations consume the same multidimensional state and, unlike the calibration loop, act only on the tutor's presentation and never on the certified record.

IV. PERFORMANCE EVALUATION

Opening paragraph (to write): state the central question — does letting a learner's self-assessment adapt the knowledge model improve it, in authentic use, without inflating certification? Preview the real-cohort-first structure and name the anti-circularity design (a graded component walled off from the tracer). Emphasize the deployment is the primary evidence, simulation the support.

A. Deployment and Study Design

WHAT GOES HERE: the credibility foundation for everything below. - Course, cohort, duration: introductory C-programming, 3 months, N = [PLACEHOLDER: 50-70] students, IRB [PLACEHOLDER]. - How the system was used in the course and that it was part of graded work. - Telemetry logged (turn_log, prediction_log, self-assessment

events). - **CRITICAL** — the walled-off independent measure: describe the graded component (proctored quiz / exam subset) that the calibration loop never ingests, and state explicitly that p_G adaptation and certification never read it. This is the design fact the whole anti-circularity argument depends on. **ARTIFACT**: a small table summarizing cohort, timeline, data volume (turns, self-assessment events, held-out items). Ethics/consent note.

B. Does the Calibration Loop Fire?

WHAT GOES HERE: the honest gate on the whole contribution. If learners don't use the slider, the mechanism is inert and nothing downstream matters, so this is reported first among results. - Slider engagement: fraction of learners who used it, frequency per learner, distribution over topics. - How many learners crossed the sustained-downgrade trigger ($N_i=3$, $EMA_i-0.3$) that actually adapts p_G (Eq. for Layer 2). - How often p_G reached its per-tier ceiling vs. adapted partially. **ARTIFACT**: histogram of self-assessment events per learner + a table of trigger/adaptation counts. Report honestly even if engagement is modest; frame low engagement as a finding, not a failure.

C. Calibration Against Held-Out Performance

WHAT GOES HERE: the centerpiece and the anti-circularity result. - Core comparison: do mastery estimates for adapted learners predict the **INDEPENDENT** held-out measure better than the un-adapted baseline (e.g., pre-adaptation estimates, or matched non-triggering learners)? - Metrics against the **EXTERNAL** measure, not the model's own outcomes: Brier score, calibration/reliability curve, AUC or point-biserial between certified/adapted status and held-out score. - Make the independence explicit in the prose: the improvement is measured against data the loop never saw, which is what rules out circularity. **ARTIFACTS**: (1) reliability diagram, adapted vs. baseline, against held-out performance; (2) table of Brier / AUC with 95Expected direction: adapted learners' estimates track external performance at least as well, and the loop reduces over-crediting for overconfident learners specifically.

D. Learning and Self-Regulation Outcomes

WHAT GOES HERE: the broader classroom signal beyond the core metric. - Relationship between loop engagement and outcomes on the held-out measure. - Whether learners' self-report calibration (slip-vs-gap / JOL agreement) improves over the semester — do their confidence judgments get more accurate? - Supporting SRL signals available in telemetry: help-seeking quality, path adherence, frustration/flow trajectories, as descriptive context. **ARTIFACTS**: outcome-vs-engagement plot; a table of SRL indicators over time (early vs. late semester). Keep causal claims cautious — observational.

E. Usability in Authentic Use

WHAT GOES HERE: whether learners found the self-assessment interaction usable and worthwhile over a full term. - Standardized instrument (e.g., SUS) + items on the slider/self-assessment specifically (was it understandable, did they trust it, did it feel useful). - Brief qualitative feedback themes. **ARTIFACT**: small table of usability scores; 2-3 representative quotes.

F. Controlled Validation via Simulated Learners

WHAT GOES HERE: the mechanism check, deliberately **AFTER** the real results and framed as isolating what the classroom cannot cleanly separate. - Synthetic learners with known ground-truth mastery, including overconfident profiles (report mastery above true), run with and without the loop. - Show the loop moves p_G in the intended direction and reduces over-crediting exactly where expected, holding everything else fixed — the clean causal version of the classroom's C result. - Optionally: sensitivity to beta, the EMA threshold, and alpha. **ARTIFACT**: bar/line plot of over-crediting with vs. without the loop across learner profiles, 95

NOTE: No LLM-as-judge subsection. That evaluates response quality (the IRL paper's concern) and does not test whether the calibration loop improves mastery estimates. Add one **ONLY** if this paper also makes a response-quality claim; otherwise it is a rung that doesn't bear weight here.

Placeholder for evaluation results.

V. CONCLUSION

Placeholder for conclusion.

REFERENCES

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